

# Causal Operator–Operand Factorization in the Residual Stream of LLMs

Matias Podeley

Independent researcher

mpodeley@gmail.com

## Abstract

How do language models represent a relational operation — *currency-of*, *capital-of* — as distinct from the entity it applies to? We separate three levels of evidence. **Representation:** a two-way ANOVA factorizes the mid-network state as  $H \approx \mu + \text{operand} + \text{operator} + \text{interaction}$ , ~90% additive across Qwen3-1.7B/8B and Gemma-2-9B — and the steering vector is *identically* the ANOVA operator component. **Causal influence:** adding  $v(\text{op\_B}) - v(\text{op\_A})$  flips the target-vs-source logit margin for all ordered relation pairs in all three models; the query token alone matches the full-prompt effect at 60× lower off-task cost, and the decisive nulls — directions rebuilt under **permuted relation labels**, and random directions **inside the operator subspace** — abolish the effect. **Behavioral sufficiency:** a state *composed* from the factorization's parts,  $\mu + \text{operand} + \text{operator}$ , patched at one position, makes the model *produce* the target answer at its own competence ceiling (52% vs. 53% clean; leave-one-out components reach 36%, and swapping the operand component redirects the answer). The interaction term is unnecessary — and the earlier failure of additive steering to generate is a **dose artifact**: at the calibrated dose the same direction generates at ceiling, while overdosing keeps inflating the margin as it pushes the state off-manifold. The structure generalizes (held-out operands, re-framed and re-lexicalized prompts, an independent animal-taxonomy domain), separates the operation from its surface realization (*language-of*  $\neq$  *demonym-of* despite both emitting "Italian"), and does not emerge for arithmetic or logic under the identical pipeline. Relational computation here is an operand plus a transplantable operator — sufficient, not merely influential, where the answer is decided.

## 1 Introduction

When a language model completes "*The currency of Italy is—*", does it represent *currency-of-Italy* as one fused thing, or as an entity plus an operation applied to it? This paper argues, causally, for the second reading — and asks the question a claim like that must answer at **three separate levels of evidence: representation** (does a stable relational direction *exist*, separable from the operand?), **causal influence** (does intervening on it systematically move the model's *relative preference* between answers?), and **behavioral sufficiency** (can the factorized parts make the model actually *produce* the target answer?). Distinguishing the levels matters because interventions that move a logit margin are routinely read as if they controlled behavior, and the gap between the two is where interpretability claims quietly overreach. Here that gap becomes the object of study — and resolves: we show it is a property of *how* the direction is applied (mode and dose), not of *what* it contains.

A line of work shows that a *task* or *relation* can be captured by a single addable vector in an LLM's residual stream (task vectors, Hendel et al., 2023; function vectors, Todd et al., 2024) and that many relations are approximately linear operators (LRE, Hernandez et al., 2024; Merullo et al., 2024). What has not been characterized is the **joint structure of operator and operand**: whether the operation *factorizes* from its argument, *generalizes*, is separable from the *word it produces*, and suffices to *drive generation* — and whether any of this holds beyond relational facts. We answer these on Qwen3 and Gemma-2, across a geographic and an animal-taxonomy domain, with arithmetic and logic as the contrast. A closing discussion (§5) offers **declension** as the organizing metaphor for the asymmetries observed.

## 2 Related work and what is new here

**Extracting relational concepts.** The closest prior work is Wang et al. (2024), who locate hidden states that express a relation separately from its subject, transplant these relational representations between subjects, and rewrite relations across 22 relation types and several models. We instead characterize the **joint geometry** of relation and entity: we quantify its additive and interaction components (the two-way ANOVA with a measured fusion term), run exhaustive all-pairs swaps against matched-norm nulls, and test whether relation directions generalize causally across held-out entities, prompt contexts, architectures, and task domains — including the domains where the factorization fails.

**Operation as an addable vector; relation as a linear operator.** Task Vectors (Hendel et al., 2023) and Function Vectors (Todd et al., ICLR 2024) extract one vector per ICL task and add/patch it to trigger the task (similarly In-Context Vectors, Liu et al., 2024; ActAdd, Turner et al., 2023) — but bundle operator *and* operand into one unfactored "task" direction, without competitive nulls. LRE (Hernandez et al., 2024) fits a full affine map per relation — a *monolithic* operator. Summing Up the Facts (Chughtai et al., 2024) decomposes factual recall into additive *circuits*; ours is a two-way ANOVA of the **residual state itself**, with a measured interaction term and a sufficiency test on the components.

**Attribute subspaces / reading position.** RAVEL (Huang et al., 2024) disentangles multiple *attributes of one entity* into subspaces (operators sharing an operand); we measure the orthogonal axis, **operator vs operand**. The operand→operator shift along the sequence restates the subject-enrichment → attribute-extraction dynamics of Geva et al. (2023) in operator/operand terms — our least novel point, included as confirmation.

**Global workspace and the Jacobian lens.** Gurnee et al. (2026) introduce the Jacobian lens and identify a mid-depth "global workspace" band. We adopt their depth band as the site of intervention; our contribution is orthogonal — the operator/operand structure of what the band *holds* — and we report a controlled null on the lens's readout advantage for our task (§4.9).

**Arithmetic and logic.** LLM arithmetic is reported as a "bag of heuristics" (Nikankin et al.,

2025) and Fourier/helical procedures (Nanda et al. 2023; Zhong et al., 2023; Kantamneni and Tegmark, 2025), with the operand *value* cleanly linear (Gurnee and Tegmark, 2024); truth is steerable (Marks and Tegmark, 2023) and logical-operator heads exist (Hong et al. 2024). No prior work reports a **cross-domain operator/operand factorization**; we provide one and show where it does not emerge.

**What is new here is the combination, not any single ingredient.** That relations admit addable vectors and linear decoders is established; what has not been done is the joint, quantified analysis: the operator  $\oplus$  operand **factorization of the residual state itself**, with the steering vector shown to *be* the ANOVA operator component; the all-pairs swap validated against **permuted-label and within-subspace nulls**; **held-out-operand generalization**; the **compositional sufficiency** test (does a state assembled from the parts *generate?*) with its dose  $\times$  position calibration; the **operation-vs-realization dissociation**; and the **cross-domain test** (each claim, its evidence, and its decisive control are tabulated at the head of §4). The declension reading is our *presentation* of these measurements, not an additional empirical claim.

## 3 Method

**Setup.** Qwen3-1.7B/8B throughout, plus Gemma-2-9B for the cross-architecture replication (§4.7); all run in bf16 on a single AMD Strix Halo APU (environment details in the supplementary repository). A relation is rendered into a template; the canonical frame is "The {op} of {a} is", and §4.5 additionally uses two phrase frames (question–answer and discourse-prefixed) that hold the {op} of {a} unit fixed while varying the surrounding frame, all ending in *is* so answer scoring is comparable. Reading position is the query token unless stated; the depth "workspace" band (38–92% of layers) is where Gurnee et al. (2026) locates the global workspace and where we build operator directions.

**Notation.** The model's state is a grid  $h_{\{\ell, t\}} \in \mathbb{R}^d$ : one residual-stream vector per layer  $\ell$  per token position  $t$ ; every measurement fixes both indices. Two reading positions recur: the **entity token** (the operand's last subword,  $t = -2$  in the canonical frame) and the **query token** (the final prompt token,  $t = -1$ , where the next-token

prediction is read; "query" means reading position, not attention's Q). Rendering the operator-operand grid through one template gives, at each  $(\ell, t)$ , a family of states  $h_{\{\ell, t\}}(o, k)$ .

**Operator direction  $\equiv$  factorization component.** For operator  $k$ ,  $v_{\ell}(k) = \text{mean}_o[h_{\{\ell, -1\}}(o, k) - \text{mean}_{\{k'\}}h_{\{\ell, -1\}}(o, k')] - \text{one direction per layer}$  of the workspace band (13 layers per model), the function-vector construction. This is *algebraically identical* to the operator main effect of the two-way factorization below ( $b_{\ell}(k) = \text{mean}_o h_{\{\ell, -1\}}(\cdot, k) - \mu_{\ell}$ ), and we verify the identity numerically (max relative deviation  $8.5 \times 10^{-8}$  across the band). Everything the paper does — steering, composing, decoding — manipulates this one object; the differences between experiments are differences in *how it is applied*, never in what is extracted.

**Factorization.** At a fixed  $(\ell, t): h = \mu + a(o) + b(k) + e(o, k)$  — operand main effect, operator main effect, and interaction, exact by construction (the interaction is the cell residual), every term carrying the  $(\ell, t)$  indices; we report the variance share of each term (Appendix A). Table 1 uses the mid-workspace layer — **layer 17 of 28 (63% depth) at 1.7B, 23 of 36 at 8B, 27 of 42 on Gemma-2-9B** — at  $t = -1$  and, as the control,  $t = -2$ .

**Swap, positions, and controls.** Efficacy of  $k_A \rightarrow k_B$  is the change in  $\text{logit}(\text{answer}_B) - \text{logit}(\text{answer}_A)$  at the query position when  $\alpha \cdot (v(k_B) - v(k_A))$  is added — over the band or a single layer, at all prompt positions or a chosen one (all / query / operand / sentence-initial control). Beyond the **matched-norm random** direction, the null battery includes directions rebuilt under **permuted relation labels** (per-operand and global — same statistics, no semantics: the decisive null), random directions **inside the operator subspace**, the correct direction at the **wrong layer** and **wrong position**, an **other-relation** direction of matched norm, and **shuffled-answer** scoring. Every intervention reports margin change, normalized margin, target rank before/after, top-1, greedy exact match (3 tokens), on-task KL, and off-task KL on unrelated WikiText — not flips alone. Answers are scored at their distinguishing first token (bare digit for arithmetic); a tokenization guard drops, per operator pair, operands whose answers collide there (4 of 12 survive on

the syncretic demonym  $\leftrightarrow$  language pairs; all 12 elsewhere).

**Composition (behavioral sufficiency).** The factorization's component vectors, computed per layer, are recombined and **replace** the residual at the query position ( $\mu + \text{operator}$ ,  $\mu + \text{operand}$ ,  $\mu + \text{operand} + \text{operator}$ ,  $\mu + \text{interaction}$ , the full donor, a norm-matched magnitude control) — plus two loophole-closers: a **leave-one-cell-out** reconstruction whose components never saw the target cell, and a **wrong-operand** composition that should (and does) redirect the answer. The generation readout ("says X") is **containment, not literal exact match**: the greedy continuation contains the gold answer case-insensitively, so *the euro*, *euros*, and *Euro* all count for *euro*; the model's own clean accuracy under the same readout is the ceiling. A generation **audit** (Appendix A) extends this: longer decodes, a five-way classification of every raw generation (target / source / right-relation- wrong-entity / wrong-relation / degraded), and a **forced-choice** score — length-normalized full- sequence log-probability among the operand's gold answers under the same intervention — which is immune to article interception and multi-token surface variants.

**Generalization and domains.** Build  $v(\text{op})$  from half the operands; test the swap on the held-out half — a genuine operator vs. interpolation among build examples. Domains: five country relations (currency, capital, language, demonym, continent  $\times$  12 countries), an independently curated **animal-taxonomy domain** (class, habitat, diet, covering  $\times$  12 animals; stable 1-to-1 answers, tokenizer-screened, three paraphrase frames), and arithmetic / comparison logic as the contrast domains.

**Statistical treatment.** The ordered swaps are **not** independent observations: they share operator directions and operands. We report cluster-bootstrap percentile intervals (10,000 replicates) at two levels: within a pair the **operand** is the resampling unit; across the paradigm the **operator** is the top-level cluster (a dyadic node bootstrap — resample operators with replacement, weight each pair by its endpoints' multiplicities, resample operands within surviving pairs). Per-operand values are released as long-form artifacts (Appendix A).

**J-space controls.** We repeat readout-geometry in the J-lens readout  $\text{unembed}(J \cdot h)$  vs the logit-lens readout  $\text{unembed}(h)$ , and re-run

efficacy with **spectrum-matched random-projection** and **permuted-vocabulary** null lenses (Appendix B).

## 4 Results

Each experiment below tests one claim against the control that could kill it:

### 4.1 Relational operators are causally manipulable — and the effect is localized and label-specific

All 20 ordered operator swaps flip the **target-vs-source logit margin** while the matched-norm random control is  $\approx 0$ . Under the operator-level cluster bootstrap (§3), the swap-random contrast is **+22.6, 95% CI [+14.0, +32.1]** at 1.7B, **+26.0 [+17.9, +32.8]** at 8B, and **+30.4 [+23.8, +34.8]** on Gemma-2-9B; the flip fraction is **1.00 [1.00, 1.00]** in all three. Per-pair operand-bootstrap CIs never cross zero, and the weakest pairs are exactly the syncretic ones (demonym  $\leftrightarrow$  language, which share their surface form; per-pair distributions in Appendix A).

**Where the vector has to land (1.7B,  $\alpha = 4$ ).** Injecting at the **query token alone** reproduces the all-position effect on the margin — at  $60\times$  lower off-task cost:

The margin effect is a **localized edit of the queried relation** — position-specific (the operand token and a structurally irrelevant token do essentially nothing; sign-correct 98/98/91% for the three effective rows vs. 4–16% for the controls) and nearly free off-task when restricted to the query position — not a global perturbation of processing. The battery replicates at 8B (query-only +32.7 vs. all-positions +33.2, at  $63\times$  lower off-task cost). (Ranks are medians, clean baseline 570; n per condition is 224 cells; bootstrap unit in §3.)

**The label is everything.** Directions rebuilt from the *same* residuals with **permuted relation labels** — preserving every statistical property of the extraction while destroying its semantics — abolish the effect: **+0.7 [-0.8, +3.9]** per operand, +0.6 global (20 redraws each; at 8B +0.15 and +0.09 against a real contrast of +26.0). A random direction **inside the operator subspace** (right subspace, wrong content) also does nothing: **+0.75 [-2.2, +3.5]** (8B: +0.43). Three further probes stay nonzero for *mechanistically informative* reasons, and decompose what the mar-

gin measures: an **other-relation** direction retains the shared  $-v(\text{source})$  component and yields  $\approx$  half the effect (+11.1 — uninstalling the source is half the margin); **shuffled-answer scoring** still moves (+21.1) because  $+v(\text{target})$  boosts the whole answer *category* (any capital city, not specifically Rome) — operand specificity lives in generation, which §4.2 measures directly; and the correct direction at the **wrong (early) layer** retains +9.8 because residual-stream additions *persist* downstream — per-layer locality is measured properly in §4.6. (Forest plot in Appendix A.)

**Dose-response, both signs.** The effect is a **signed causal axis**: the operator-specific shift is monotone in  $\alpha$  through zero and *reverses* with the direction — +22.6 at  $\alpha = +4$  vs. **-21.0 at  $\alpha = -4$** , an almost exact mirror (8B: +26.0 vs. -24.5), saturating at both ends (the positive branch higher, +29 vs. -15 raw, because the clean margin already favors the source: a floor) — while the random control's shift ( $\approx +6$ ) is flat in  $|\alpha|$ , which is why every headline number is a swap - random contrast. Off-task cost grows with dose and breadth (7.9  $\rightarrow$  21 nats/token band-wide vs. 0.29 at the query token). What dose does to *generation* is the subject of §4.2. (Curves in Appendix A.)

### 4.2 A state composed from the factorization's parts generates the answer

At the default dose, additive steering flips every *margin* but rarely makes the model *say* the target answer (0.4–2.2% exact match), whereas replacing the query-position residual with a **real donor activation** (same operand, target relation) does — 51%, at the model's clean accuracy of 53%. Is the donor carrying something the direction lacks — operand binding, a non-additive interaction, lexical or execution signals? We decompose the donor by the exact factorization and patch partial reconstructions at the query position (every variant includes  $\mu$ : the hook *replaces* the state, so a variant must be a plausible full-magnitude residual). 1.7B, 224 (pair, operand) cells:

**The donor carries nothing the additive parts don't.** The purely additive composition  $\mu + \text{operand} + \text{operator}$  generates at the donor's own level (51.8% vs. 50.9%) — the model's competence ceiling — while the interaction term alone does nothing (7.1%, vs. 4.5% for the magnitude control). Two controls close the loopholes. A **leave-one-cell-out** reconstruction, whose operand and operator components were

| claim                                                                                                                                                                                                      | main evidence                                                                                                                                                                                                                                                                                                                                                             | decisive control                                                                                                                                                                                                                                                                                  | scope                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| a stable operator direction exists<br>it causally controls the margin<br>the effect is localized<br>the operator generalizes<br>the parts <b>suffice to generate</b><br>the factorization is domain-scoped | ANOVA ~90% additive; $v(\text{op}) \equiv$ main effect (§4.3)<br>all-pairs swaps flip, 3 models (§4.1)<br>query token $\approx$ all positions, $60\times$ lower KL (§4.1)<br>held-out operands; re-framed, re-worded prompts (§4.5)<br>composed patch $\approx$ donor $\approx$ ceiling; calibrated steering (§4.2)<br>animals replicates; arithmetic/logic do not (§4.8) | reading-position control<br><b>permuted labels</b> $\rightarrow 0$ ; subspace-random $\rightarrow 0$<br>operand / sentence-initial positions $\approx 0$<br>random $\approx 0$ ; transfer $\approx$ within<br>leave-one-out; wrong-operand; magnitude; interaction<br>held-out fails off-relation | 9 relations, 3 models<br>relative preference<br>both Qwen3 scales<br>relations + animals<br>greedy, this setup<br>our setup (§4.8) |

Table 1: The paper’s claims, each with its main evidence, the control that could kill it, and its scope.

| position, scope     | $\Delta\text{margin}$ [95% CI] | rank(tgt) | says target | KL off      |
|---------------------|--------------------------------|-----------|-------------|-------------|
| all, band           | +28.9 [+16.2, +39.9]           | 431       | 0.4%        | 18.4        |
| <b>query, band</b>  | +28.7 [+16.1, +39.8]           | 368       | 2.2%        | <b>0.29</b> |
| query, one layer    | +13.7 [+7.7, +18.3]            | <b>69</b> | 9.8%        | 0.04        |
| operand, band       | +1.8 [+1.0, +2.5]              | 3194      | 0.4%        | 0.26        |
| wrong (t = 0), band | +2.3 [-0.2, +4.8]              | 194       | 3.6%        | 1.56        |

Table 2: Where the vector must land (1.7B,  $\alpha = 4$ ): position  $\times$  layer scope, with the metrics flips hide. Ranks are medians over 224 cells; clean-baseline rank 570.

| patched state                                       | says target  | says source | says other   | rank(target), median | top-1 |
|-----------------------------------------------------|--------------|-------------|--------------|----------------------|-------|
| magnitude control ( $\mu$ + norm-matched random)    | 4.5%         | 33.5%       | 0.0%         | 696                  | 0%    |
| interaction only ( $\mu$ + inter)                   | 7.1%         | 35.3%       | 0.0%         | 166                  | 0%    |
| operand only ( $\mu$ + operand)                     | 7.6%         | 37.5%       | 0.0%         | 54                   | 0%    |
| operator only ( $\mu$ + operator)                   | 20.1%        | 14.7%       | 1.8%         | 71                   | 0%    |
| <b>operator + operand (<math>\mu</math> + both)</b> | <b>51.8%</b> | 13.4%       | 1.8%         | <b>2</b>             | 31.7% |
| operator + operand, <b>held-out cell</b>            | <b>35.7%</b> | 18.8%       | 1.8%         | 5                    | 17.9% |
| operator + <b>wrong</b> operand                     | 20.1%        | 11.6%       | <b>34.4%</b> | 206                  | 0%    |
| full donor ( $\mu$ + both + inter)                  | 50.9%        | 12.5%       | 1.8%         | <b>2</b>             | 31.2% |

Table 3: Decomposing the donor activation (1.7B, 224 cells; clean greedy ceiling 53%): a state composed from the factorization’s additive parts makes the model emit the target answer as well as the real donor does; the interaction term is not needed. "Held-out cell" rebuilds the components without ever seeing the target cell; "wrong operand" redirects the answer to the swapped operand.

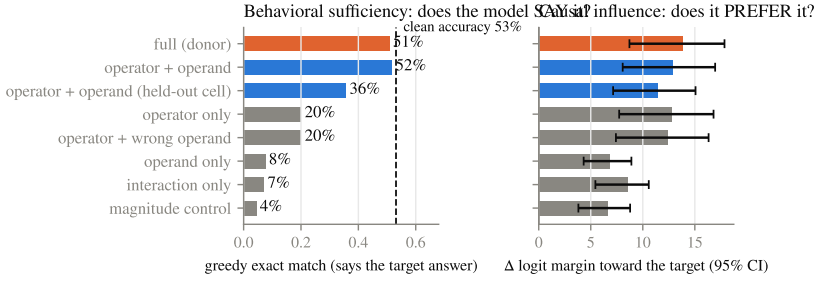
built *without ever seeing the target cell*, still generates at 35.7% — composition, not leakage. And swapping in a **wrong operand’s** component redirects the answer to *that* operand (34.4% says-other vs. 1.8% baseline): the state is compositional in exactly the way the factorization claims — the operator component selects the relation, the operand component selects whose answer. At **8B every number sharpens**: composed 62.1% vs. donor 65.2% (clean ceiling 68%), leave-one-out **50.4%**, wrong-operand redirect **48.2%**, interaction-only 3.6%, magnitude control 2.2%.

**Sufficiency by steering, at the calibrated dose.** Since  $v(\text{op})$  is the operator component (§3), adding  $\alpha \cdot (v(B) - v(A))$  at the query position of one layer lands, at  $\alpha = 1$ , on  $\mu + \text{operand} + \text{operator}(B) + \text{interaction}(A)$  — the composed state up to a wrong interaction term. The dose sweep

confirms it behaviorally: generation follows an **inverted-U peaked exactly at the on-manifold dose** — 38.4% at  $\alpha = 1$  for a single layer, **51.3% at  $\alpha \approx 0.1$  for the 13-layer band** (band additions accumulate, so the band’s on-manifold dose is  $\approx 1/13$  per layer; the same peak doses reproduce at 8B: 45.1% and 56.2%) — while past the peak the margin keeps climbing (+28.9 at  $\alpha = 4$ ) as generation collapses. The earlier "steering moves margins but cannot generate" result was an **overdose artifact**, not an information deficit: the factorized representation is **compositionally sufficient**, by replacement and by calibrated addition. (At a single layer,  $\mu + \text{operator}$  alone already generates at 44.6% — unpatched layers still carry the operand, so the injection point is local and operand identity distributed.)

**The generation audit** (Appendix A) closes the metric loopholes directly. Classifying every raw

A state composed from the factorization's parts generates the answer — the fusion term is not needed



Query-position patch at the workspace band, 1.7b, 224 (pair, operand) cells. Every variant includes  $\mu$  (the hook REPLACES the residual, so a variant must be a plausible state). "full (donor)" =  $\mu$  + operand + operator + interaction is the real donor activation by construction. "held-out cell" rebuilds the operand and operator components WITHOUT ever seeing the target cell. CIs: operator-level cluster bootstrap.

Figure 1: Decomposition ladder: greedy exact match and margin change for each partial reconstruction of the donor activation. The additive composition matches the full donor at the model's clean ceiling; interaction-only and magnitude controls do nothing; the held-out-cell reconstruction and the wrong-operand redirect close the leakage and specificity loopholes.

k = 8 generation: the calibrated dose produces the *fluent, correct* answer (59.4% target vs. 0.9% source; composed patch 57.6% — both at the clean prompt's own 58.5%), while at the overdose **94.6% of generations are degraded token loops** by manual inspection — category-correct, fluency-destroyed, not surface variants the metric missed. A **forced-choice** readout (full-sequence log-prob among the operand's gold answers, under the intervention) then dissociates information from fluency: the target wins at **80.4% even at the overdose** (87.5% calibrated, 84.4% donor patch, 2.7% clean). Overdosing destroys the generation manifold, not the encoded preference.

### 4.3 The representation factorizes into operator $\oplus$ operand

Does the state literally decompose into "the thing" plus "the question asked about it"? Two-way ANOVA of  $h_{\{\ell, \tau\}}(o, k)$  at the mid-workspace layer (L17/28 at 1.7B, L23/36 at 8B; §3) — one decomposition per reading position:

~90% additive; operand and operator components occupy distinct subspaces (principal angles 41–85° at the query token — though near-orthogonality is the high-dimensional default, so the load-bearing evidence is the variance split and the causal results). Since the steering vector is numerically this ANOVA's operator main effect (§3), the direction that flips margins (§4.1), the component that composes into a generating state (§4.2), and the term that dominates this split are **one vector seen three ways**. The operator-dominant representation **emerges along the sequence**: the entity enters operand-dominant and is re-marked by

its queried role at the query position (the reading-position control rules out template echo).

(A PCA of the 60 workspace states, showing the operand-organized cloud at the entity token reorganizing into operator clusters at the query token, is in Appendix A.)

### 4.4 Operation $\neq$ realization (syncretism)

*language-of* and *demonym-of* emit the **same word** (Italian) yet are **distinct operator directions**; the swap between them is weak precisely because they share their output word. An **exponent-free operator marker** built from the shared-output pairs ( $\text{mean}[h(\text{language}) - h(\text{demonym})]$ , the output word cancelled by construction) still installs the relation causally (1.7B:  $-3.6 \rightarrow +12.5$ ): the operator representation is not an output-word representation (§5 reads this as syncretism).

(Operator-direction cosines and the exponent-free-marker bar chart are in Appendix A.)

### 4.5 Operators generalize: held-out operands, prompt frames, wordings — and a second domain

**Held-out operands.**  $v(\text{op})$  built on 6 operands and applied to the 6 held-out still flips **20/20** swaps — a transferable operator, not interpolation: contrast **+20.0 [+10.8, +29.5]** at 1.7B, **+22.8 [+14.0, +31.0]** at 8B, flip fraction 1.00 at both.

**Frames and wordings.** Rendering every relation in three paraphrase frames and testing every build→test combination: on 1.7B **all 9 combinations flip 20/20** (180/180), *frame-invariant to two decimals* (+22.6 on all three test frames); at 8B



| read position | operand          | operator         | interaction (fusion) |
|---------------|------------------|------------------|----------------------|
| query token   | 5% / 6%          | <b>86% / 82%</b> | 9% / 13%             |
| entity token  | <b>59% / 55%</b> | 31% / 34%        | 9% / 11%             |

Table 4: Two-way ANOVA of the workspace state: variance shares by read position (1.7B / 8B).

100/100 with the same invariance. Stronger: the direction survives changing the *wording of the relation itself* — re-lexicalizing every relation (*currency of*  $\rightarrow$  *money used in*, ...) and transferring both ways gives **40/40 at both scales**, with transfer  $\approx$  within-formulation (+22.59 vs. +22.62 at 1.7B). Clean baselines shift across frames and wordings; the operator's causal effect does not: it is a property of the **relation**, not its wording.

**A second, non-geographic domain.** Every signature replicates on an independently curated animal-taxonomy domain (class / habitat / diet / covering  $\times$  12 animals; 1.7B): **12/12** ordered swaps flip (+18.1 [+13.5, +22.8]); held-out operands **12/12** at +18.0 — transfer indistinguishable from within; cross-frame **72/72**, again frame-invariant to two decimals; permuted-label and subspace nulls at zero (-0.5 and +0.2, CIs spanning 0); the same reading-position signature (operand 79% at the entity token, operator 62% at the query token); and even the same quantitative margin decomposition (the other-relation probe retains  $\approx$  half the effect in both domains: 47% vs. 49%). Two honest caveats: the animal factorization carries more interaction (22% vs. 9% — informative in §4.8), and the generation-level readout is floor-limited here (the model's own clean greedy accuracy on these multi-token answers is 6%), so in this domain sufficiency is attested by margins and rank, not exact match.

#### 4.6 Where the operator is represented is not where it acts

A per-layer sweep (1.7B; direction built **and** injected at each single layer) separates three notions the band framing conflates. *Trivial readability*: a held-out-operand classifier decodes the operator at **100% from every layer, including layer 0** — the operator's token is in the prompt, so decodability locates nothing. *Representation*: the ANOVA operator share peaks **early** (92% at 19% depth), decaying to 57%. *Causal leverage*: the single-layer

swap contrast is  $\approx 0$  early and peaks **late** (+21.3 [+13.8, +28.2] at 89% depth — one layer there reproduces the full band's +22.6). The two informative curves peak **70 depth-points apart** (8B: 23% vs. 89% — the same profile): where the operator is most cleanly factorized is not where injecting it most moves the output. This explains the wrong-layer probe of §4.1 (early additions act by persisting into later layers) and diagnoses the mid-band layer of §4.1–4.2 as far from optimal (L17: +6.8 vs. L24: +21.3): being *readable*, being *the representation*, and being *causally potent* are three different layer profiles of one direction. (Figure in Appendix A.)

#### 4.7 Cross-architecture replication (Gemma-2-9B)

The entire pipeline transfers unchanged to **Gemma-2-9B** — a different pretraining corpus, tokenizer (SentencePiece; answer single-token coverage 0.95), and architecture family (soft-capped logits, GQA). Every qualitative signature replicates, with the strongest effects of the three models: all-pairs contrast **+30.4 [+23.8, +34.8]** (20/20 flips), held-out **+26.6 [+20.6, +33.2]**, operator variance 84.8% with **6.8%** interaction, and the exponent-free marker  $-3.0 \rightarrow +8.5$  (full three-model table in Appendix A). Raw logit units are not directly comparable across architectures (Gemma-2 soft-caps its final logits), so cross-model comparison should lean on flip rates and variance shares, which match or exceed Qwen3's: the factorization is not an idiosyncrasy of one model family.

#### 4.8 Domain scope: a second retrieval domain replicates; arithmetic and logic do not emerge

If the factorization were an artifact of prompting, it should appear wherever the prompt has the same shape. It does not. Extending the identical pipeline to arithmetic (+,  $\times$ , -) and comparison-logic operators, on 1.7B, with the animal domain

(§4.5) as the second retrieval column:

| domain           | op var | fusion | held-out           | swap/rand   |
|------------------|--------|--------|--------------------|-------------|
| <b>countries</b> | 86%    | 9%     | <b>20/20</b>       | +21 / ~0    |
| <b>animals</b>   | 62%    | 22%    | <b>12/12</b>       | +19 / +0.7  |
| arithmetic       | 55%    | 23%    | 2/6 ( $\leq$ rand) | +0.4 / +0.3 |
| logic            | 33%    | 34%    | 2/6 ( $\leq$ rand) | +1.2 / +0.2 |

The gradient replicates and sharpens at **8B** (arithmetic held-out 1/6; logic interaction 45%, held-out **0/6**): relational  $\gg$  arithmetic  $>$  logic at both scales. The animals row adds the decisive nuance: its interaction share (22%) is essentially arithmetic's (23%), yet its held-out transfer is perfect and arithmetic's is at chance — **what separates domains is whether the operator direction transfers, not how much fusion the state carries**. This is a **domain contrast, strictly scoped**: the factorization observed in relational retrieval does not emerge for arithmetic **under our models, tokenization choices, prompts, and intervention procedure** — a regime where 1.7B/8B arithmetic competence is itself limited, digit tokenization restricts the grid, and the operation may be computed distributively or iteratively rather than retrieved. We claim the contrast, not a theorem about algorithmic computation.

**Reconciling a contrary report.** Christ et al. (2025) find that **add-N** operators (a fixed addend N over one number) *do* generalize. The cuts differ: add-N varies a linear numeric *parameter* — effectively an operand on the number line (Gurnee & Tegmark 2024) — while our  $+$   $\times$   $-$  cut varies the *function*. Running both cuts on Qwen3 places add-N between relations and  $+$   $\times$   $-$  (held-out 5/12  $\cdot$  6/12), and its directions are by far the most collinear of the three families (76% of the operator-set variance on one line — a number line, not a set of operations). Their positive and our negative coexist without contradiction; the full table is in Appendix A.

#### 4.9 This is causal structure, not a J-space readout advantage

This project began as a replication of the J-space readout claim of Gurnee et al., 2026, whose "workspace" band we intervene on through-out. Under matched controls — including a spectrum-matched random projection — that readout showed **no advantage** over the logit lens on any of four comparisons (Appendix B): the structure here is causal organization, not a readable

subspace.

## 5 Discussion

**Modes of intervention, and what a margin measures.** What looked like a gap between influencing a preference and executing an operation was a gap between *ways of applying the same vector*: a large added multiple moves margins while pushing the state off-manifold; a calibrated multiple, or replacement with the composed state, generates at ceiling (§4.2). The null battery likewise decomposes the margin — half uninstalling the source, half installing the target *category*, operand specificity visible only in generation (§4.1). Both points travel: margin shifts at one fixed dose measure category-level preference under a possibly off-manifold intervention, and negative generation results do not, by themselves, show the information is absent. The three-level discipline — representation, influence, sufficiency, with sufficiency tested by *composition* — is the transferable method here.

**Relational computation as declension.** The measurements invite a morphological reading, offered as an organizing metaphor rather than a claim. In a declining language a noun's role is marked by a case ending on a stem: the same stem takes different endings, the same ending applies across stems (§4.5), two cases can share a surface form without being the same case (**syncretism** — *language-of* and *demonym-of*, §4.4), stem and ending compose with a small residue of **fusion** (§4.3, behaviorally inert), and the composed stem-plus-ending is itself a usable form (§4.2). The limits: unlike morphological case, the marking is continuous, graded, and **layer-dependent** — a trajectory, not a static affix (§4.6) — and arithmetic and logic do not decline (§4.8). The honest formulation is that entities acquire **transient relational configurations partially comparable to grammatical case**, not that the model implements a case grammar.

## 6 Conclusion

An LLM represents a relational operation at three verifiable levels. The operator **exists** as a stable direction — identically the factorization's operator component. It **causally controls** the answer margin — every swap flips, localized at the query token, vanishing under permuted labels. And the factorization is **behaviorally sufficient**: a state composed from its additive parts, with no inter-



action term, makes the model produce the target answer at its own competence ceiling, by replacement or by calibrated addition. The structure generalizes to unseen operands, rewordings, a second domain, and a second architecture; it separates the operation from the word that realizes it; it does not emerge for arithmetic or logic under our setup. The gap between shifting a preference and executing an operation, where it appeared, was a property of the intervention, not the representation.

## 7 Limitations

**Scale and family:** 1.7–9B models from two decoder-only families; no  $\geq 30$ B model; relational linearity holds for a subset of relations even in-domain (cf. LRE  $\sim 48\%$ ). **Sufficiency is scoped:** "generates" means greedy exact match over three tokens against the model's own clean accuracy as ceiling, on short single-query prompts; composition and dose  $\times$  position are at 1.7B (8B replication in the artifacts); the leave-one-out composition reaches 36%, not the full-grid 52% — some of the composed state's precision does come from seeing the cell. **Coverage:** five country + four animal relations, English-only; three context frames and a full re-lexicalization (§4.5); other entity types and languages open. **Arithmetic:** single-digit regime (BPE), at scales where arithmetic competence is itself limited (§4.8). **Reading-position result** restates known subject-enrichment  $\rightarrow$  attribute-extraction dynamics. **No J-space claim** (§4.9, Appendix B).

## References

- Miranda Anna Christ, Adrián Csiszárík, Gergely Becsó, and Dániel Varga. 2025. The structure of relation decoding linear operators in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:2510.26543, spotlight.
- Bilal Chughtai, Alan Cooney, and Neel Nanda. 2024. Summing up the facts: Additive mechanisms behind factual recall in llms. *arXiv preprint arXiv:2402.07321*.
- Mor Geva, Jasmijn Bastings, Katja Filippova, and Amir Globerson. 2023. Dissecting recall of factual associations in auto-regressive language models. In *Proceedings of EMNLP*. ArXiv:2304.14767.
- Wes Gurnee, Nicholas Sofroniew, Jack Lindsey, et al. 2026. Verbalizable representations form a global workspace in language models. *Transformer Circuits Thread*.
- Wes Gurnee and Max Tegmark. 2024. Language models represent space and time. In *International Conference on Learning Representations (ICLR)*. ArXiv:2310.02207.
- Roe Hendel, Mor Geva, and Amir Globerson. 2023. In-context learning creates task vectors. In *Findings of the Association for Computational Linguistics: EMNLP 2023*. ArXiv:2310.15916.
- Evan Hernandez, Arnab Sen Sharma, Tal Haklay, Kevin Meng, Martin Wattenberg, Jacob Andreas, Yonatan Belinkov, and David Bau. 2024. Linearity of relation decoding in transformer language models. In *International Conference on Learning Representations (ICLR)*. ArXiv:2308.09124.
- Jing Huang, Zhengxuan Wu, Christopher Potts, Mor Geva, and Atticus Geiger. 2024. Ravel: Evaluating interpretability methods on disentangling language model representations. In *Proceedings of ACL*. ArXiv:2402.17700.
- Subhash Kantamneni and Max Tegmark. 2025. Language models use trigonometry to do addition. *arXiv preprint arXiv:2502.00873*.
- Sheng Liu, Haotian Ye, Lei Xing, and James Zou. 2024. In-context vectors: Making in-context learning more effective and controllable through latent space steering. In *International Conference on Machine Learning (ICML)*. ArXiv:2311.06668.
- Samuel Marks and Max Tegmark. 2023. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *arXiv preprint arXiv:2310.06824*.
- Jack Merullo, Carsten Eickhoff, and Ellie Pavlick. 2024. Language models implement simple word2vec-style vector arithmetic. In *Proceedings of NAACL*. ArXiv:2305.16130.
- Yaniv Nikankin, Anja Reusch, Aaron Mueller, and Yonatan Belinkov. 2025. Arithmetic without algorithms: Language models solve math with a bag of heuristics. In *International Conference on Learning Representations (ICLR)*. ArXiv:2410.21272.
- Eric Todd, Millicent L. Li, Arnab Sen Sharma, Aaron Mueller, Byron C. Wallace, and David Bau. 2024. Function vectors in large language models. In *International Conference on Learning Representations (ICLR)*. ArXiv:2310.15213.
- Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. 2023. Activation addition: Steering language models without optimization. *arXiv preprint arXiv:2308.10248*.
- Zijian Wang, Britney Whyte, and Chang Xu. 2024. Locating and extracting relational concepts in large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*. ArXiv:2406.13184.

Ziqian Zhong, Ziming Liu, Max Tegmark, and Jacob Andreas. 2023. The clock and the pizza: Two stories in mechanistic explanation of neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:2306.17844.

## A Reproducibility, stimuli, and implementation details

**Checkpoints and hardware.** Qwen/Qwen3-1.7B, Qwen/Qwen3-8B, google/gemma-2-9b, all bf16 on a single AMD Strix Halo APU (128 GB unified memory, ROCm PyTorch). Seeds default to 0 and affect only the random controls; direction building is deterministic.

**Stimuli.** Operands (12 countries): Italy, Japan, Russia, China, France, Germany, Brazil, Egypt, India, Mexico, Sweden, Turkey. Relations: currency, capital, language, demonym ("word for a person from"), continent; every (country, relation) cell has a gold one-word answer (e.g. Italy → euro, Rome, Italian, Italian, Europe); the full grid ships in `data/relations.json`, the lexical reformulations in `data/relations_lex.json`, and the paraphrase frames in the dataset metadata.

**Answer scoring and selection.** Answers are scored at the first token of " " + answer. Within each ordered operator pair, operands whose two answers collide at that token are dropped (4 of 12 survive on the syncretic demonym↔language pairs; all 12 elsewhere). Whole-word single-token coverage: 0.92 (Qwen3 BPE), 0.95 (Gemma SentencePiece).

**Directions, interventions, and controls.**  $v(op)[layer]$  is the mean over (operand, frame) cells of the op's residual minus that cell's mean over ops, computed at ~25 evenly spaced source layers restricted to the 38–92% depth band. Interventions add  $\alpha \cdot (v(to) - v(from))$  at every position of the band ( $\alpha = 4$  unless swept). The matched-norm control draws one Gaussian vector per layer and rescales the band-concatenated stack to exactly the injected difference's norm. Clean mean margins  $\logit(to) - \logit(from)$  are  $\approx -7$  to  $-8$  across models and frames; per-operand values for every experiment are released as long-form artifacts (`results/ablation/*_long.parquet`).

### Method schematic.

**The animal-taxonomy domain.** Operands (12): shark, eagle, frog, snake, bee, whale, owl,

lizard, cow, salmon, penguin, deer. Relations: class, habitat, diet, covering; every cell has a stable 1-to-1 answer (e.g. shark → fish, ocean, carnivore, scales), screened so all four answers are first-token distinct per operand (single-token coverage 0.62 — multi-token answers like *carnivore* depress the greedy ceiling but not the first-token margin metric); three paraphrase frames mirror the country domain; grid in `data/animals.json`, authoring gate in `build_op_datasets.py`.

**The generation audit (full table, §4.2).** 1.7B relations, 224 cells/condition,  $k = 8$  greedy, raw texts persisted in the artifact; forced choice = length-normalized sequence log-prob among the operand's gold answers under the same intervention:

Containment classification can also over-count (2–5% of cells contain both target and source — "the Swedish krona" contains demonym-"Swedish" as a modifier); worst-case reclassification moves target rates by  $\leq 5$  points.

**Cross-architecture replication (full table, §4.7).**

### Per-pair distributions.

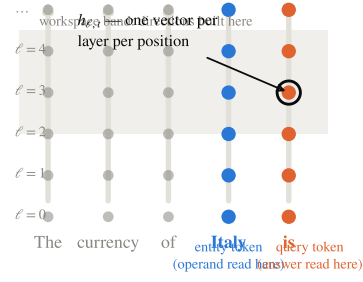
**The null battery (full results, 1.7B relations,  $\alpha = 4$ ).** Contrast vs. matched-norm random with operator-level cluster-bootstrap 95% CIs; stochastic nulls pool 20 redraws:

**The add-N comparison (moved from §4.8).** Held-out generalization and operator-set collinearity for the three operator families:

Qwen3 BPE restricts add-N to single-digit results (5 operands), so its generalization is weak in absolute terms; the claim is the ordering, not the magnitude.

**Experiment drivers.** `op_core.py` (directions  $\equiv$  ANOVA components, swaps, position-restricted hooks, metric bundle, cluster bootstraps), `operator_paradigm.py`, `operator_factorize.py`, `operator_templates.py`, `operator_lexical.py`, `op_dose.py` (signed  $\alpha$  sweep), `op_minimal.py`, `op_patch_decomp.py` (composition ladder), `op_positions.py` (position  $\times$  scope  $\times$  dose), `op_nulls.py` (the battery above), `op_layer_sweep.py` (decodability vs. causal effect per layer), `op_geometry_dump.py`; `verify_numbers.py` recomputes every number in this paper from the persisted artifacts.

The state is a grid: layers  $\times$  positions



One object, applied three ways

at each band layer  $\ell$ , average the query-token states:

$$v_\ell(\text{op}) = \text{mean}_o [h_{\ell, -1}(o, \text{op}) - \text{mean}_k h_{\ell, -1}(o, k)]$$

$\equiv$  the ANOVA operator main effect  $b_\ell(\text{op})$  (verified: dev.  $10^{-8}$ )

**steer** add  $\alpha(v_\ell(B) - v_\ell(A))$  at the query token  $\rightarrow$  margin flips; generates at calibrated  $\alpha$

**compose** replace  $h_{\ell, -1}$  with  $\mu_\ell + a_\ell(o) + b_\ell(k) \rightarrow$  generates at the clean ceiling

**decode** classify / ANOVA the same states  $\rightarrow$  factorization, layer profiles

Prompt  $\rightarrow$  token  $\times$  layer grid of residual states  $\rightarrow$  per-layer operator direction by averaging at the query token  $\rightarrow$  applied by addition (steering), by replacement (composition), or read out (ANOVA).

Figure 2: Method in one schematic: the model's state is a token-by-layer grid of residual vectors; the entity token and the query token are the two reading positions; averaging query-token states over operands yields one operator direction per layer — identically the ANOVA operator main effect — which is then applied by addition (steering), by replacement (composition), or read out (ANOVA).

| condition                                 | target | source | other | degraded | forced choice |
|-------------------------------------------|--------|--------|-------|----------|---------------|
| clean                                     | 3.6%   | 58.5%  | 1.3%  | 36.6%    | 2.7%          |
| patch composed ( $\mu$ +operand+operator) | 57.6%  | 11.6%  | 0.9%  | 29.9%    | 76.3%         |
| patch donor                               | 57.6%  | 9.8%   | 0.8%  | 31.7%    | 84.4%         |
| add band $\alpha=4$ (overdose)            | 0.4%   | 0.0%   | 4.9%  | 94.6%    | 80.4%         |
| add band $\alpha=0.1$ (calibrated)        | 59.4%  | 0.9%   | 1.3%  | 38.4%    | 87.5%         |
| add one layer, query, $\alpha=1$          | 40.6%  | 21.9%  | 2.2%  | 35.3%    | 76.3%         |

| measure                                       | Qwen3-1.7B               | Qwen3-8B                | Gemma-2-9B                                |
|-----------------------------------------------|--------------------------|-------------------------|-------------------------------------------|
| all-pairs swap contrast                       | +22.6 [+14.0, +32.1]     | +26.0 [+17.9, +32.8]    | <b>+30.4 [+23.8, +34.8]</b>               |
| flips (swap > 0)                              | 20/20                    | 20/20                   | <b>20/20</b>                              |
| held-out-operand contrast                     | +20.0 [+10.8, +29.5]     | +22.8 [+14.0, +31.0]    | <b>+26.6 [+20.6, +33.2]</b>               |
| operator variance @ query                     | 86%                      | 82%                     | <b>84.8%</b>                              |
| interaction (fusion)                          | 9%                       | 13%                     | <b>6.8%</b>                               |
| exponent-free marker (clean $\rightarrow$ +v) | -3.6 $\rightarrow$ +12.5 | -2.9 $\rightarrow$ +8.1 | <b>-3.0 <math>\rightarrow</math> +8.5</b> |

Table 5: Cross-architecture replication: every signature reproduces on Gemma-2-9B (logit units are not comparable across architectures; compare flip rates and variance shares).

## B The J-space readout comparisons

Four readout comparisons on Qwen3-1.7B (8B where noted) asked whether the Jacobian-lens readout exposes more than the logit lens: bridge-entity pass@k (a tie, drifting toward the logit lens at 8B); a surface-form logit difference with real competitors (logit lens matches or wins in every depth band); a grammatical-number probe with Hewitt–Liang selectivity (number decodable at ceiling from *all* lenses — including a **spectrum-matched random projection**, the decisive control); and the concept-plane trajectory that motivated the project (the J-lens path is longer on random planes too — a generic magnitude effect, not a concept channel). A permuted-vocabulary lens

control behaves accordingly. The 1.7B readout-geometry difference (cases reorganizing by output form in J-space) does not replicate at 8B and is not relied on. Full tables live in the project's working log.

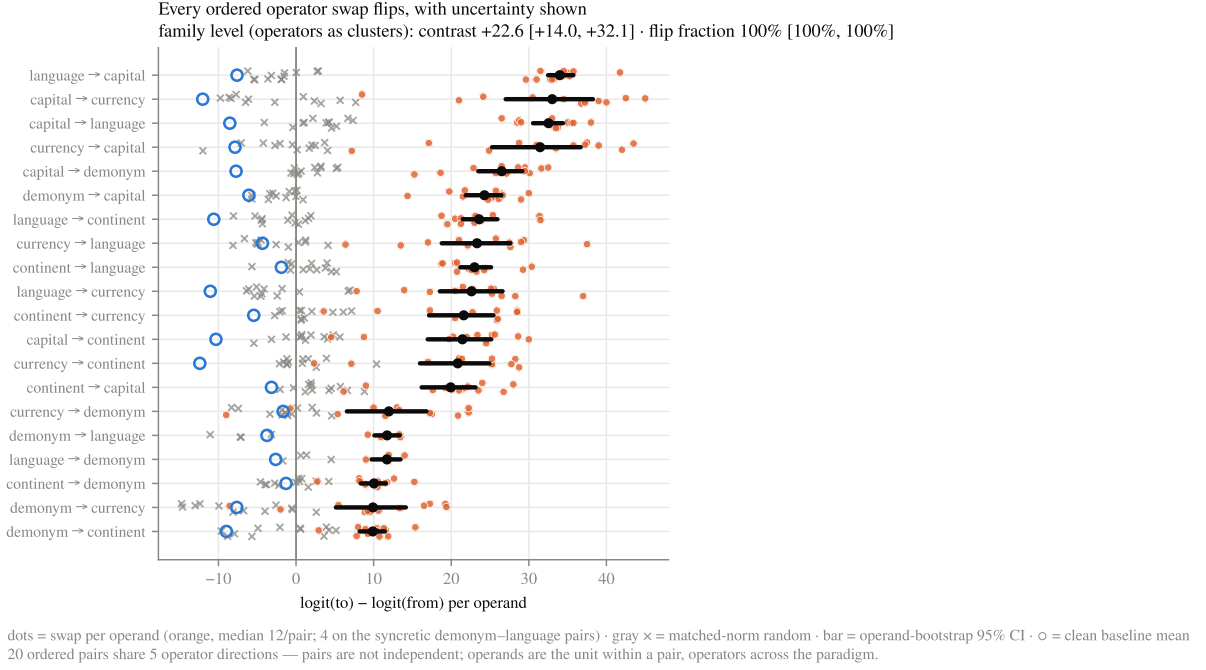


Figure 3: Per-pair distributions: swap values per operand with operand-bootstrap 95% CIs, matched-norm random control, and clean baselines. No CI crosses zero; the weakest pairs are exactly the syncretic ones (demonym↔language and their neighbours).

| condition                     | contrast [95% CI]     | flip fraction |
|-------------------------------|-----------------------|---------------|
| real swap                     | +22.62 [+14.0, +32.1] | 1.00          |
| permuted labels (per-operand) | +0.68 [-0.8, +3.9]    | 0.45          |
| permuted labels (global)      | +0.57 [-2.1, +3.0]    | 0.35          |
| random in operator subspace   | +0.75 [-2.2, +3.5]    | 0.45          |
| wrong layer (early band)      | +9.77 [+5.2, +13.3]   | 1.00          |
| other-relation direction      | +11.09 [+6.7, +17.1]  | 0.80          |
| shuffled answers              | +21.12 [+11.2, +32.2] | 1.00          |

Table 6: The null battery (1.7B,  $\alpha = 4$ ). Semantic nulls – permuted relation labels and random directions inside the operator subspace – abolish the effect; structural probes remain nonzero for mechanistically explained reasons (§4.1).

| cut                         | held-out generalization (1.7B / 8B) | operator-set collinearity (top-1 var) |
|-----------------------------|-------------------------------------|---------------------------------------|
| relations (5 operations)    | 20/20 · 20/20                       | 0.39 (spread paradigm)                |
| add-N (operator = addend)   | 5/12 · 6/12 (+0.10 / +0.05)         | 0.76 (most 1-D / number-line)         |
| + × − (operator = function) | 2/6 · 1/6 (-0.43 / -0.55)           | 0.64                                  |

Table 7: Reconciliation with the add-N cut: the linear-parameter family sits between relations and true two-operand arithmetic.

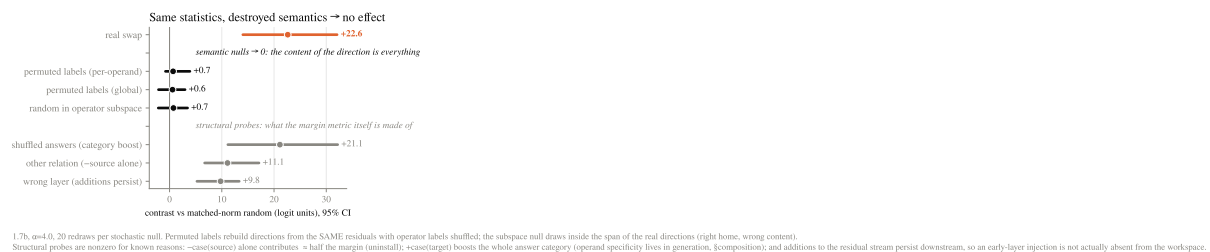
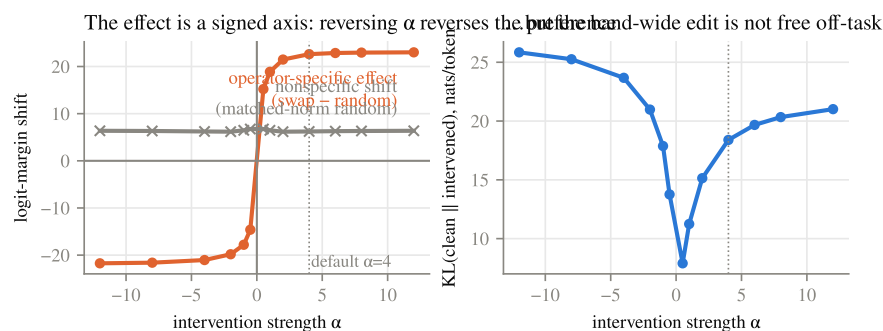


Figure 4: Forest plot of the null battery: semantic nulls (permuted labels, subspace-random) collapse to zero; structural probes remain nonzero for mechanistically explained reasons (source-uninstall component, category-level boost, additive persistence across layers).



Right: per-token KL on unrelated WikiText with the hook active at every position of the workspace band — the intervention is answer-surgical, not distribution-surgical.

Figure 5: Dose-response with both signs: the operator-specific effect is monotone through zero and reverses with the direction's sign, saturating at both ends, while the matched-norm random control is flat; per-token KL on unrelated text grows with dose.

## Where operand and operator live

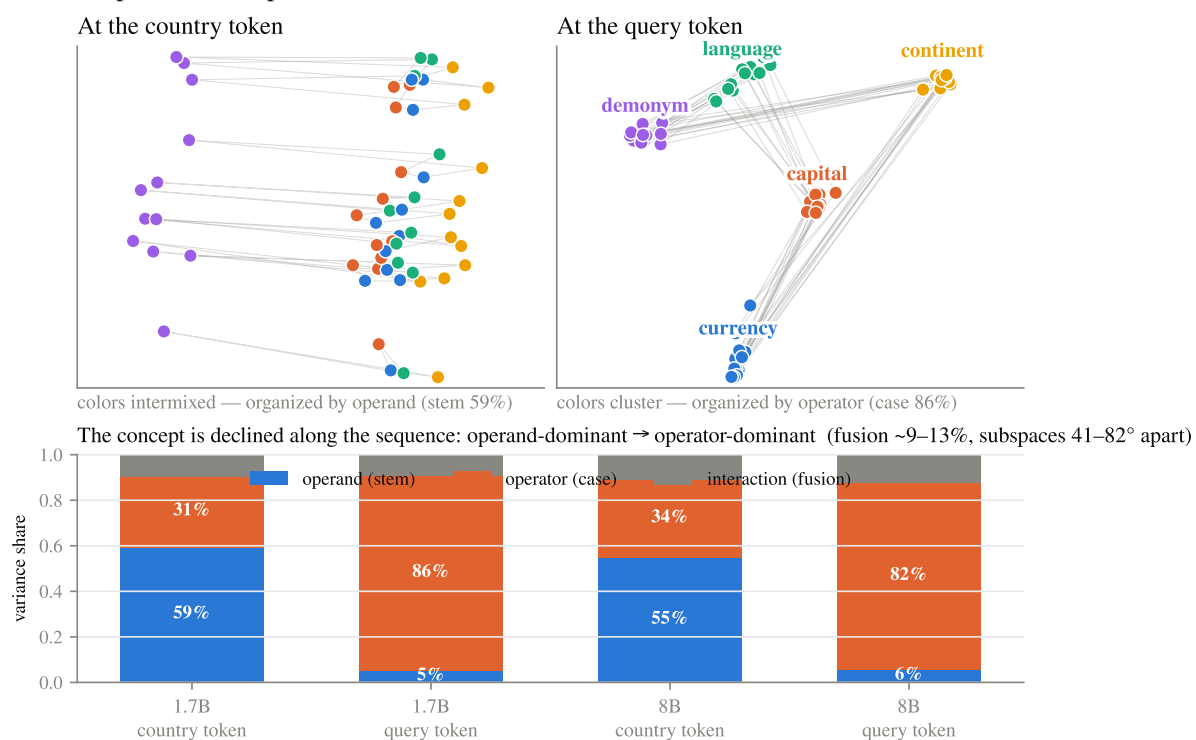


Figure 6: PCA of the 60 workspace vectors  $H[\text{operand}, \text{operator}]$ , colored by operator. At the country token the operators intermix (operand-organized, operand 59%); at the query token they separate into clean operator clusters (operator-organized, operator 86%). The faint web links each country's five operator-forms — short at the country token, splayed at the query token. Bottom: the variance split, 1.7B and 8B.

## Operation $\neq$ realization: the case is separable from the word

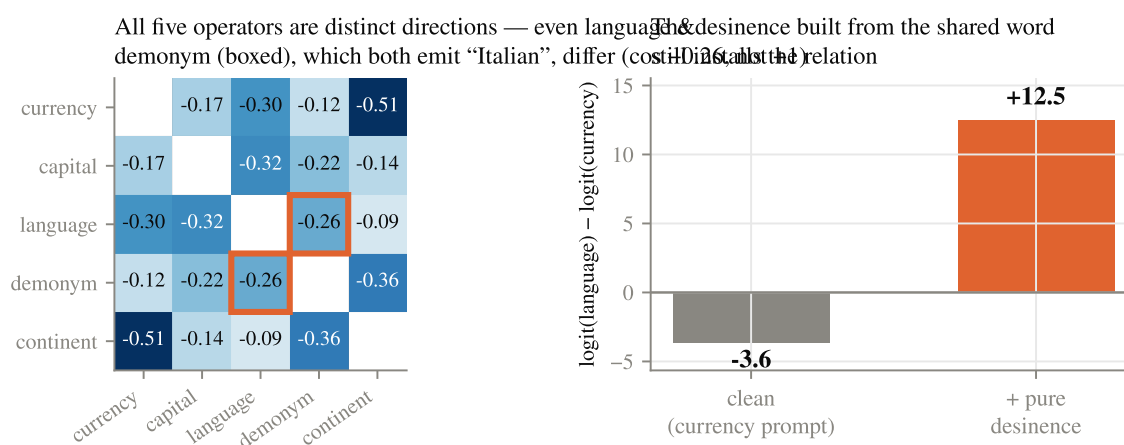
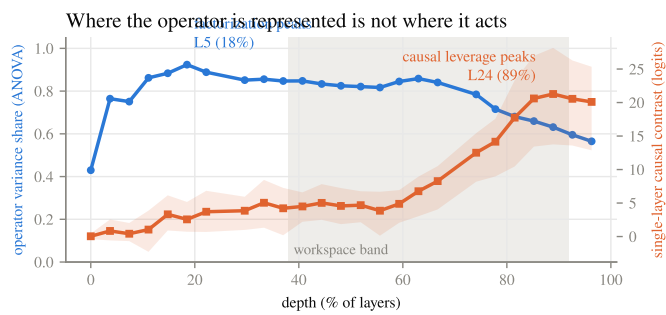


Figure 7: Left: operator-direction cosines — all five operators are distinct directions (no off-diagonal near  $\pm 1$ ); language and demonym, boxed, both emit “Italian” yet differ (cos -0.26). Right: the exponent-free marker, built where the two share an output word, still installs the relation causally ( $-3.6 \rightarrow +12.5$ ).





1.7b. Blue: operator variance share of the two-way ANOVA at that layer (query position). Orange: all-pairs swap contrast when the direction is built AND injected at that single layer (shaded = 95% CI). Held-out-operand decodability is 100%–100% at every layer — the operator token is present in the prompt, so mere readability locates nothing; these two curves do.

**Figure 8: Layer sweep: the ANOVA operator share peaks early (92% at 19% depth) while the single-layer causal contrast peaks late (+21.3 at 89% depth); held-out-operand decodability is at ceiling at every layer.**